Vancouver Business Licences Explorer

Overview

The project uses the City of Vancouver Business Licences GeoJSON dataset to analyze business patterns at two levels: individual licences and neighborhood-level business composition. This report summarizes the necessary findings from A1, A2, A3, and B4. Geographic and operational groupings were identified using K-means, DBSCAN, and PCA, and interactive neighbourhood comparison is supported by a Streamlit app.

A1 -Data Acquisition and Cleaning

There were 204,215 records and 27 columns in the raw GeoJSON. After removing the dictionary-based geo_point_2d field from the duplicate check, no exact duplicate rows were discovered. The analysis kept the 167,147 licences with status'Issued’ because they most accurately reflect authorized business activity. Records with invalid coordinates were eliminated for geographic clustering, leaving 85,934 usable business locations.

Cleaning item	Result / decision
Raw dataset	204,215 rows, 27 columns
Issued licences retained	167,147
Valid geocoded issued licences	85,934
Business subtype missing	89.46%
Fee paid missing	36.93%
Coordinates missing	49.76%
Postal code missing	46.62%
Local area missing	1.33%

Nearly nine out of ten values in the business subtype field were missing, so it was not used. Instead, the more comprehensive businesstype field was combined into 23 common categories that accounted for approximately 85% of issued licenses; the remaining categories were grouped as 'Other'.

To lessen the impact of extreme values, employee and fee variables were log transformed, and the median value of $277 was used to fill in any missing license fees. Because local area coverage was significantly stronger than postal code coverage, it was chosen for neighborhood analysis.

A2 -Location-Only Clustering

DBSCAN was used to find dense local patterns and spatial outliers, and A2 clustered businesses using only standardized latitude and longitude K-means was used to create broad citywide partitions.

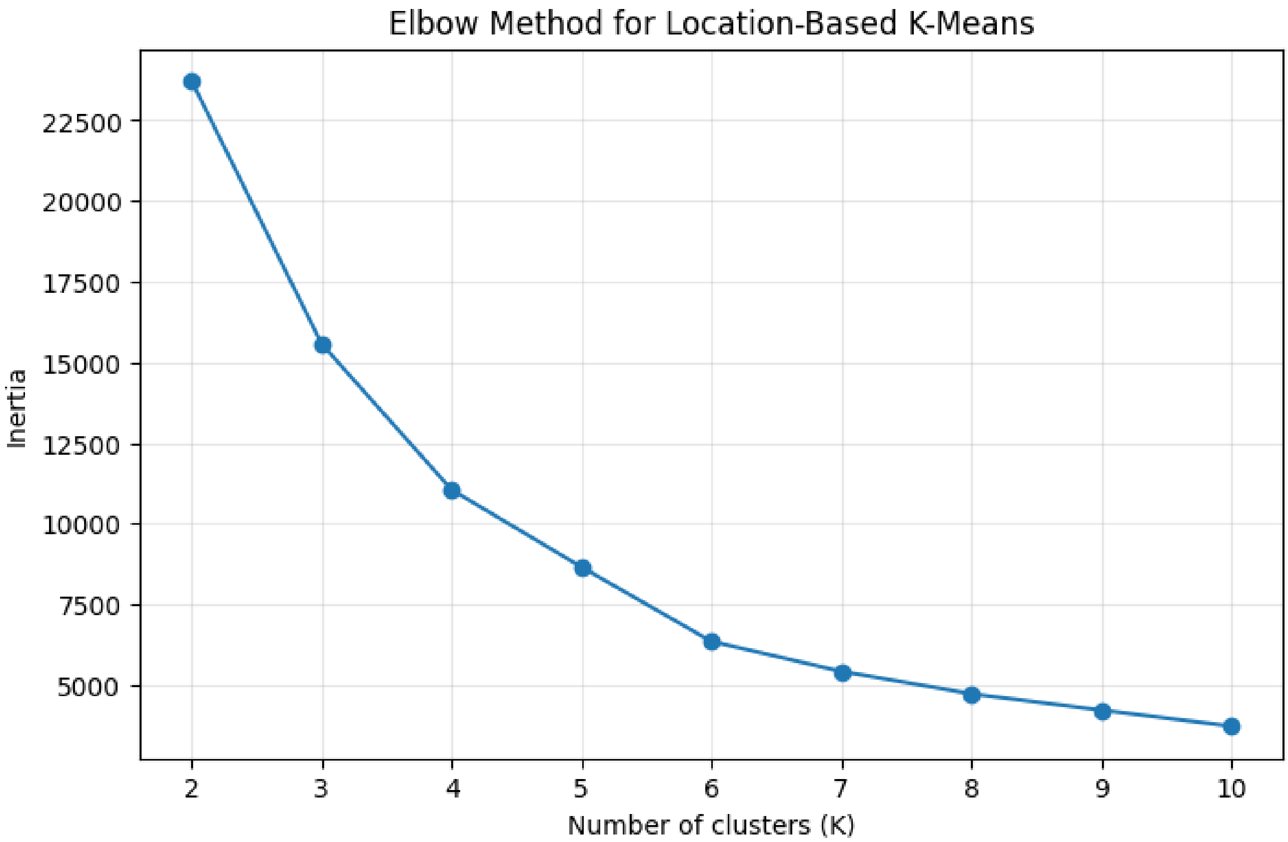

Figure 1. Elbow method for location-based K-means.

The selection of K=5 as a practical balance between model fit and interpretability is supported by the inertia curve's sharp decline from K=2 to roughly K=5, after which the improvement becomes noticeably smaller.

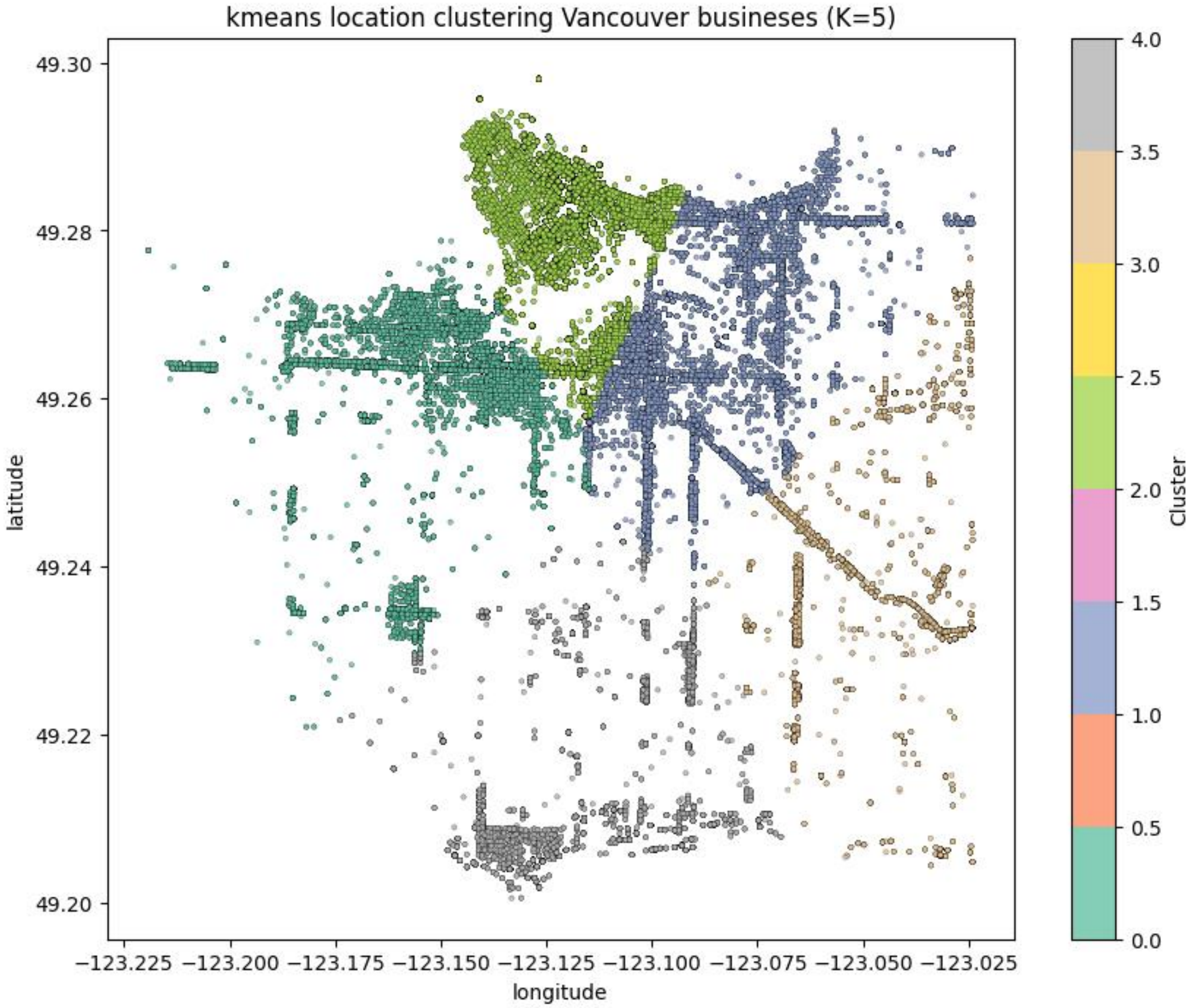

Figure 2. K-means clustering of Vancouver business locations (K=5).

Vancouver is clearly divided geographically by the five K-means clusters, which roughly divide the city's western, central, northern, eastern, and southern regions. While the smaller southern clusters have fewer licenses, the largest cluster, which encompasses the dense central and east central business network, has 38,729 businesses. K-means creates an easy-to-understand citywide segmentation by allocating each point to a single cluster and favouring compact areas.

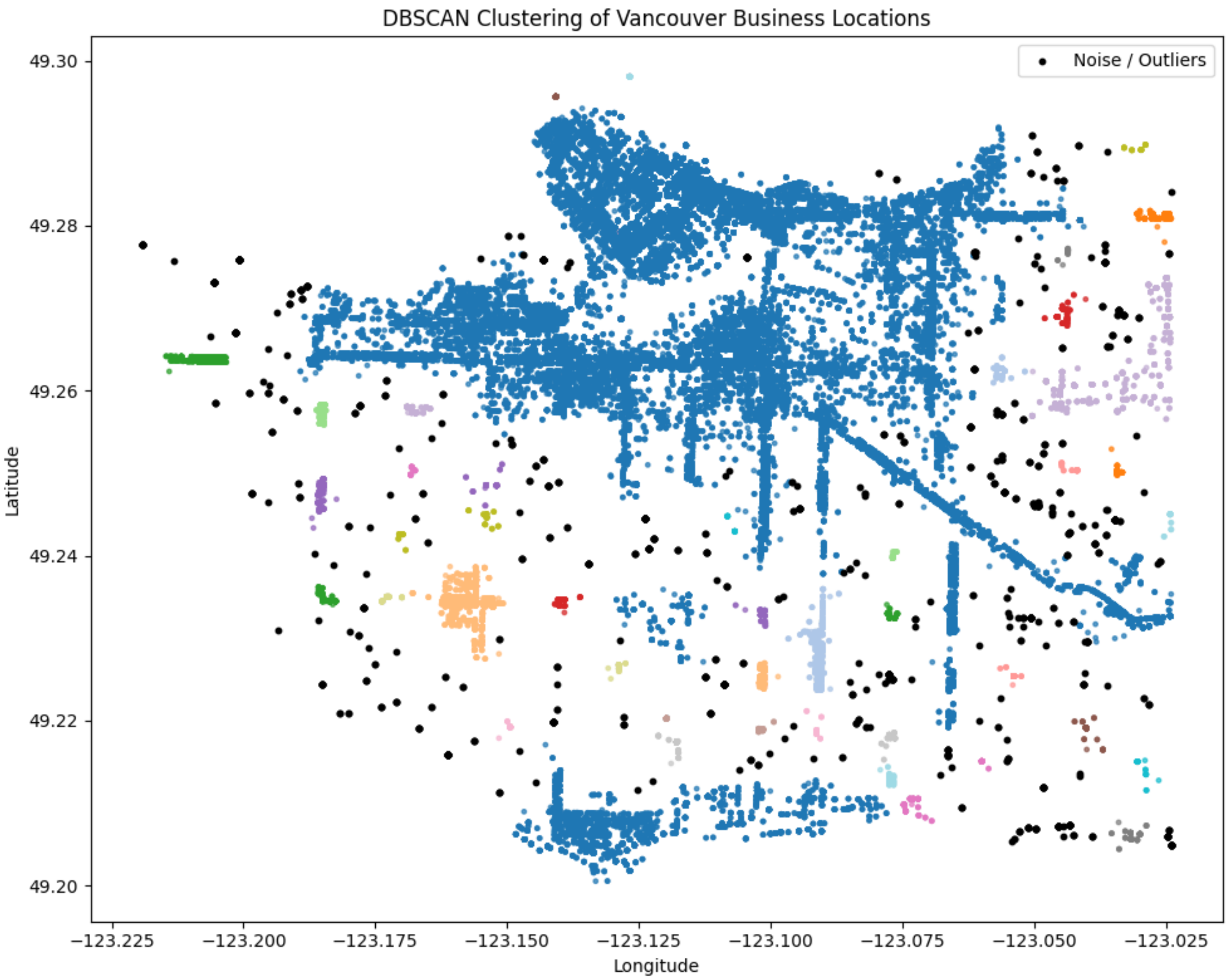

Figure 3. DBSCAN clustering with noise points shown in black.

DBSCAN generated 865 noise points, numerous smaller local clusters, and a single dominant connected cluster. Because Vancouver businesses are dispersed along continuous commercial corridors, which allow nearby dense areas to link together, a large, connected cluster most likely formed. The black points represent businesses that did not have enough nearby neighbours to meet the density requirement.

The smaller coloured groups could be concentrated business zones, shopping centres, or remote neighbourhood centres. DBSCAN is more effective than K-means at identifying local hotspots and outliers, but its output is less appropriate for a straightforward citywide regional division.

A3 -Feature-Based Clustering and PCA

Log transformed employee count and license fee, license duration, cyclic issue month, revision number, and consolidated business type were among the feature set that A3 used to group businesses based on operational characteristics rather than location. Business type was one-hot encoded, and numerical variables were standardized.

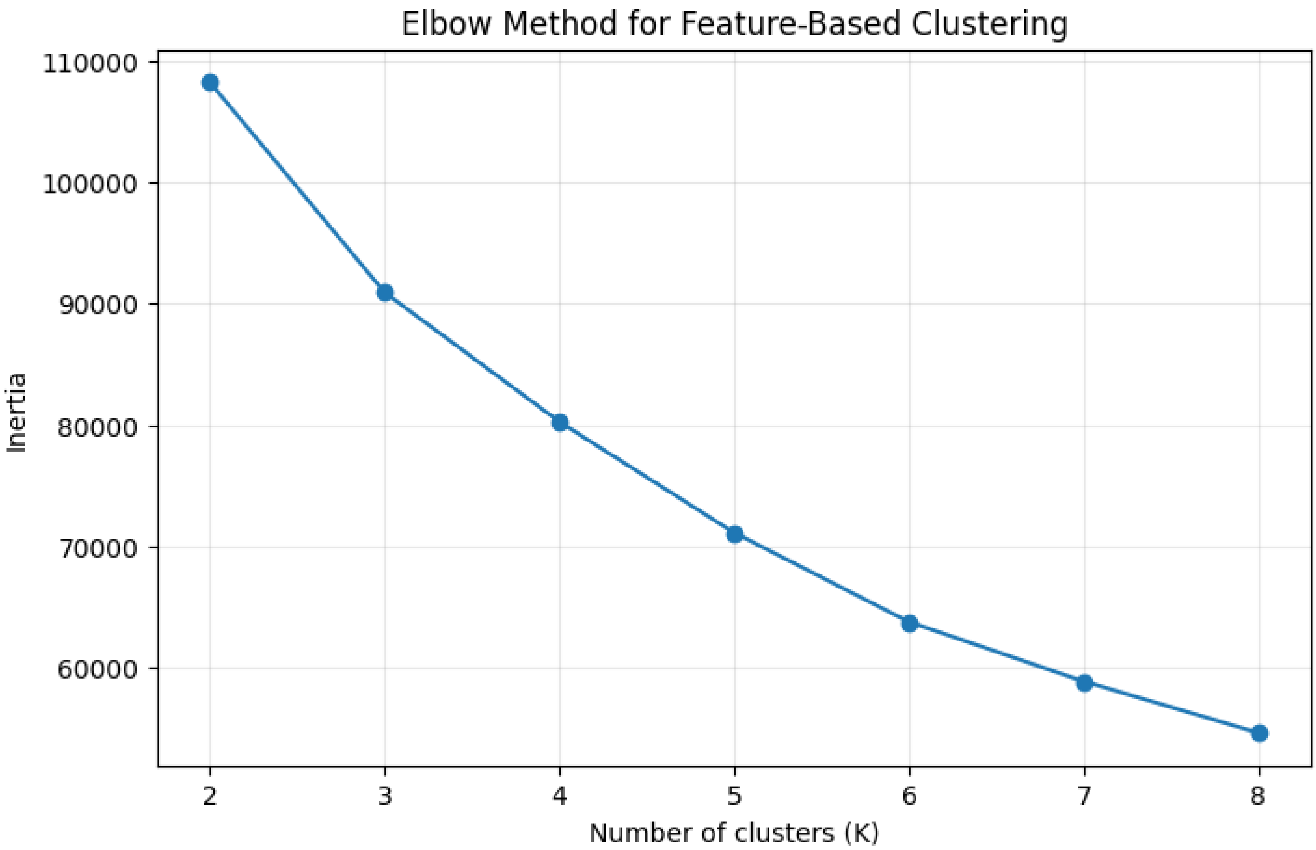

Figure 4. Elbow method for feature-based clustering.

There is no perfectly sharp choice for K because the feature based elbow curve is smoother than the location based curve. But around K = 5, the curve starts to flatten, making five clusters a reasonable and understandable option.

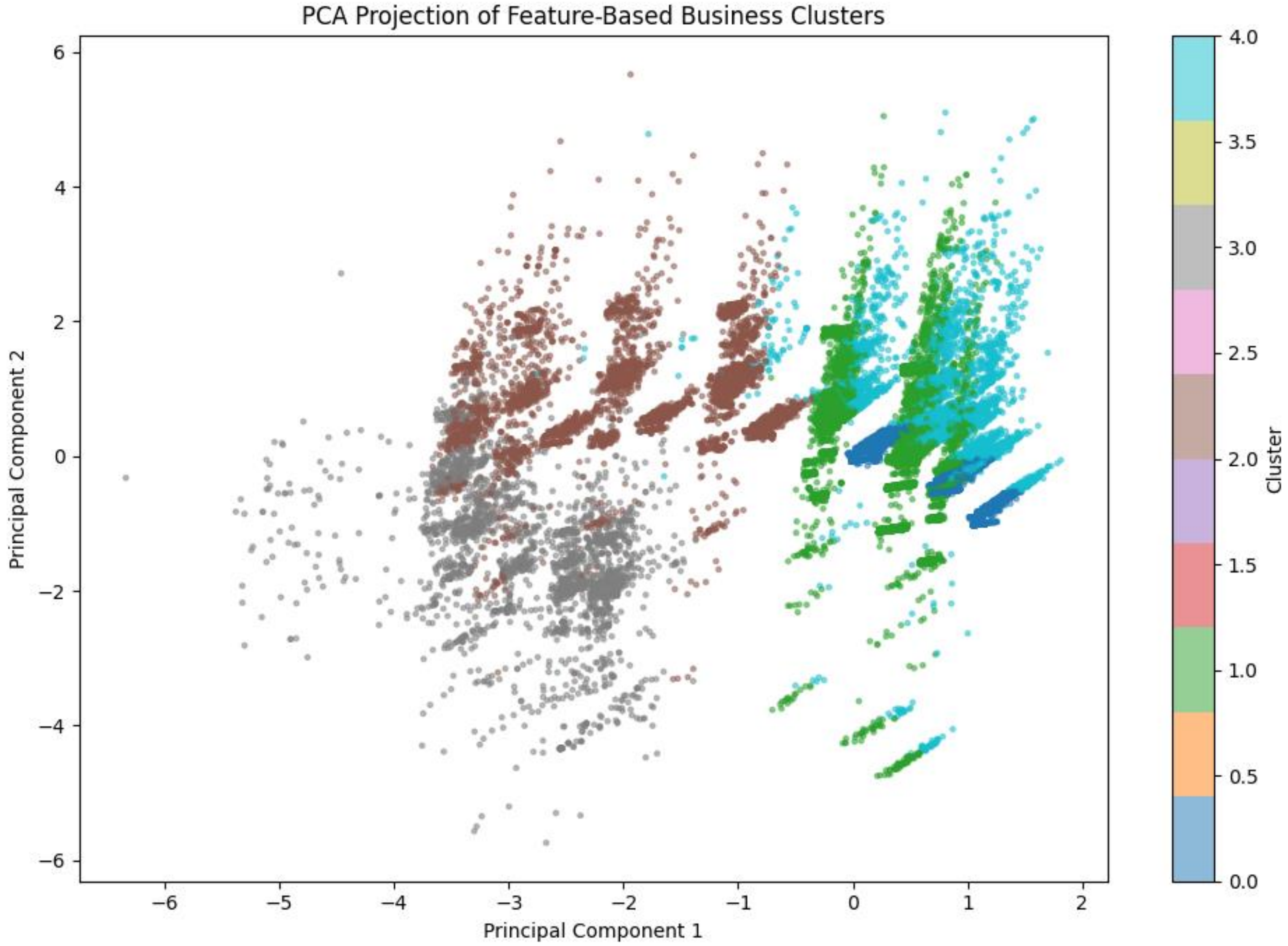

figure 5. Two-dimensional PCA projection of the five feature-based clusters.

The PCA plot clearly separates the five operational clusters; Cluster 4 stands out as the large-business group with the highest average employee and fee values, while Cluster 3 has the shortest average license duration and the lowest average fee. The first two principal components account for 43.03% of the total variance, despite some overlap. While Cluster 1 has the most businesses and nearly no revisions, Cluster 0 is distinguished by a significantly higher average revision number.

Because businesses are categorized based on how they operate rather than where they are located, these results differ from A2. Therefore, if two businesses in the same neighbourhood have different sizes, industries, fees, durations, or license histories, they may belong to different feature-based clusters.

Cluster	**Businesses**	**Main profile**
0	37,933	higher revisions (avg. 10.02); long licence duration
1	65,780	largest cluster; very low revisions; many rentals
2	23,583	moderate revisions and shorter licence duration
3	13,900	shortest duration and lowest

		average fee
4	25,795	largest employers and highest average fee

B4 -Neighbourhood Cluster Membership and Interpretation

After eliminating UBC, which had just 31 records, and 'Out of Town' which is not a Vancouver neighbourhood, a 22-by-24 composition matrix was produced for Part B, where the unit of analysis was changed from individual businesses to neighbourhoods. A neighbourhood is represented by each row, and the percentage of businesses that belong to a consolidated business type is represented by each column. This makes it possible to fairly compare regions with wildly disparate numbers of businesses.

The user can instantly view updated cluster membership in both PCA and geographic views by changing K in the Streamlit application. While areas with stronger restaurant, retail, financial, or professional-service profiles may form separate clusters, neighbourhoods with high shares of long-term rentals and short-term rental operators may group together. Downtown is expected to differ from primarily residential neighbourhoods because it contains a more diverse concentration of business support, finance, consulting, and retail activity. The main conclusion is that business similarity can transcend geographic boundaries. For example, two distant neighbourhoods may cluster together if their industry proportions are similar, but nearby neighbourhoods may split apart if their commercial profiles are different.

Conclusion

The analysis demonstrates that the clustering results are highly dependent on the features and algorithm employed. K-means produced the most distinct broad geographic division of Vancouver, whereas DBSCAN identified isolated areas, dense corridors, and local hotspots. The neighborhood-level Streamlit app showed that comparable business ecosystems can exist in various parts of the city, and feature-based K-means found significant operational profiles that were independent of geography. When combined, the approaches offer complementary perspectives on the locations of businesses, their operations, and the neighbourhoods with similar commercial structures.

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